Rhulani Osborn Manganyi
Brightlearn
Data Analytics
Exercise number 3 (case statements)
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Question 1: classifying each product by price into the three tiers

```sql
--- Question 1: Classify each product by price into three teirs
SELECT product_name,
       price,
       CASE
           WHEN price > 1000 THEN 'Expensive'
           WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
           WHEN price < 100 THEN 'Budget'
           END AS price_category
FROM exercise.case.products;
```

Add parameter

Table

	product_name	price	price_category
1	Laptop	1200	Expensive
2	Phone	800	Mid-range
3	Keyboard	45	Budget
4	Monitor	300	Mid-range
5	Mouse	25	Budget

Question 2: Label each order by its value

```sql
--- Question 2: Label each order by its value into three categories highe value, mediun value and low value
SELECT customer_name,
       amount,
       CASE
          WHEN amount >= 1000 THEN 'High Value'
          WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
          WHEN amount < 500 THEN 'Low Value'
          END AS order_value_category
FROM exercise.case.orders;
```

Add parameter

Table

	customer_name	amount	order_value_category
1	Alice	150	Low Value
2	Bob	560	Medium Value
3	Charlie	999	Medium Value
4	Diana	45	Low Value
5	Ethan	1200	High Value

Question 3: categorise each employee's position level using both department and salary

```sql
--- Question 3: categorise each employee's position level using both department and salary
SELECT emp_name,
       department,
       salary,
       CASE
          WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
          WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'
          ELSE 'Staff'
          END AS employee_level
```

Add parameter

Table

	emp_name	department	salary	employee_level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experienced HR
3	Mark	IT	75000	Staff
4	Lucy	Finance	95000	Staff
5	Tom	HR	55000	Staff

Question 4: Assign each student a letter grade based on their score

```sql
SELECT student_name,
       score,
       CASE
          WHEN score >= 90 THEN 'A'
          WHEN score BETWEEN 80 AND 89 THEN 'B'
          WHEN score BETWEEN 70 AND 79 THEN 'C'
          WHEN score BETWEEN 60 AND 69 THEN 'D'
          WHEN score < 60 THEN 'F'
          END AS grade
```

Add parameter

Table

	student_name	score	grade
1	Anna	92	A
2	Ben	76	C
3	Cara	59	F
4	David	83	B
5	Ella	68	D

Question 5: Label delivery performance based on the time taken

```sql
--- question 5: label delivery performance based on the time taken
SELECT delivery_id,
     delivery_time_minutes,
     CASE
        WHEN delivery_time_minutes <= 30 THEN 'Fast'
        WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
        WHEN delivery_time_minutes > 60 THEN 'Late'
        END AS performance
FROM exercise.case.deliveries;
```

Add parameter

Table

	delivery_id	delivery_time_minutes	performance
1	1	45	On Time
2	2	80	Late
3	3	30	Fast
4	4	65	Late
5	5	100	Late

Question 6: Convert the numeric priority into a readable label

```sql
SELECT issue_type,
     priority,
     CASE
        WHEN priority = 3 THEN 'High'
        WHEN priority = 2 THEN 'Medium'
        WHEN priority = 1 THEN 'Low'
        END AS priority_label
FROM exercise.case.tickets;
```

Add parameter

Table

	issue_type	priority	priority_label
1	Login issue	1	Low
2	Sever down	3	High
3	Slow system	2	Medium
4	Email error	2	Medium
5	Password reset	1	Low

Question 7: Calculate attendance percentage and classify the result

```sql
--- Question 7: Calculate attendance percentage and classify the result
SELECT student_id,
(days_present/total_days)*100 AS attendance_percentage,
      CASE
          WHEN (days_present/total_days)*100 >= 90 THEN 'Excellent'
          WHEN (days_present/total_days)*100 BETWEEN 75 AND 89 THEN 'Good'
          WHEN (days_present/total_days)*100 < 75 THEN 'Needs Improvement'
          END AS attendance_status
       FROM exercise.case.attendance;
```

Add parameter

Table

	student_id	attendance_percentage	attendance_status
1	1	90	Excellent
2	2	60	Needs Improvement
3	3	96	Excellent
4	4	50	Needs Improvement
5	5	100	Excellent

Question 8: Label the stock status of each product

```sql
--- question 8: classify products based on their stock levels
SELECT product_id,
       stock_qty,
   CASE
       WHEN stock_qty = 0 THEN 'Out of Stock'
       WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
       WHEN stock_qty > 5 THEN 'In Stock'
       END AS stock_status
   FROM exercise.case.product_inventory;
```

Add parameter

Table

	product_id	stock_qty	stock_status
1	1	5	Low Stock
2	2	0	Out of Stock
3	3	25	In Stock
4	4	10	In Stock
5	5	3	Low Stock

Question 9: Classify each class by the number of enrolled students

```sql
--- question 9: classify each class based on the number of enrolled students
SELECT subject,
       enrolled_students,
       CASE
           WHEN enrolled_students >= 25 THEN 'Large'
           WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
           WHEN enrolled_students < 10 THEN 'Small'
           END AS class_size_category
   FROM exercise.case.classes;
```

Add parameter

Table

	subject	enrolled_students	class_size_category
1	Math	30	Large
2	English	25	Large
3	Science	15	Medium
4	Art	5	Small
5	History	20	Medium

Question 10: Combine payment method and amount to decide whether a discount applies

```sql
--- question 10: apply a discount flag based on the payment method
SELECT *,
       CASE
            WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'
            ELSE 'Not Eligible'
             END AS discount_eligibility
   FROM exercise.case.payments;
```

Add parameter

Table

	payment_id	amount	payment_method	discount_eligibility
1	1	50	Card	Not Eligible
2	2	200	Cash	Eligible for Discount
3	3	150	Card	Not Eligible
4	4	75	PayPal	Not Eligible
5	5	300	Cash	Eligible for Discount